VM networking modes (NAT, NAT Network, Host-only, Bridged): when to use each, and why vulnerable labs must never be bridged.

VM Networking Modes: NAT, NAT Network, Host-only and Bridged

What is VM Networking?

Virtual Machine (VM) Networking refers to the process of connecting virtual machines to each other, the host computer, or external networks using virtual network adapters.

Virtualization software such as VMware Workstation and Oracle VirtualBox provides different networking modes to control how virtual machines communicate.

The four common networking modes are NAT, NAT Network, Host-only, and Bridged.

1. NAT (Network Address Translation)

NAT allows a virtual machine to access the internet through the host computer's network connection while keeping the VM behind a virtual NAT gateway.

How it works:

- The VM receives a private IP address.
- Outbound traffic passes through the NAT gateway.
- The VM can access websites, download packages, and install updates.
- External devices generally cannot initiate direct connections to the VM unless port forwarding or another route is configured.

Example: A Kali Linux VM uses NAT to download cybersecurity tools and update packages.

When to use: When a VM requires internet access for browsing, downloading tools, or performing software updates.

2. NAT Network

NAT Network allows multiple virtual machines to communicate with each other on the same private virtual network while also providing internet access through NAT.

How it works:

- Multiple VMs are connected to the same NAT Network.

- VMs can communicate with one another using their private IP addresses.
- The NAT gateway provides outbound internet connectivity.
- External devices generally cannot directly access the VMs without additional configuration.

Example: Kali Linux and an Ubuntu VM are connected to the same NAT Network for network scanning and security testing.

When to use: When multiple virtual machines need to communicate with each other and also require internet access.

Note: NAT Network is a distinct networking option in VirtualBox. VMware provides similar functionality through its virtual networking configurations.

3. Host-only Networking

Host-only networking creates a private virtual network between the host computer and the connected virtual machines.

How it works:

- The VM can communicate with the host computer.
- Multiple VMs connected to the same host-only network can communicate with each other.
- The VM cannot directly access the internet through the host-only adapter.
- It provides an isolated environment for security testing, provided no additional routing or forwarding is enabled.

Example: A Kali Linux VM and a Metasploitable VM are connected using host-only networking to practice penetration testing safely.

When to use: When performing ethical hacking, vulnerability assessment, malware analysis, or penetration testing in an isolated laboratory.

4. Bridged Networking

Bridged networking connects a virtual machine directly to the same physical network as the host computer.

How it works:

- The VM behaves like a separate device on the physical network.
- It can receive an IP address from the physical network's DHCP server.
- The VM can communicate with other devices on the LAN, subject to network controls.
- Other devices may also be able to communicate directly with the VM.

Example: A virtual machine connected in bridged mode receives an IP address from the home router and communicates with other computers on the same WiFi network.

When to use: When testing network services that require the VM to behave like an independent physical device on an authorized LAN.

Difference Between VM Networking Modes

Basis	NAT	NAT Network	Host-only	Bridged
Internet access	Yes	Yes	No, by default	Yes, if physical network permits
VM-to-VM communication	Limited by configuration	Yes, on the same NAT Network	Yes, on the same host-only network	Yes, if LAN permits
Host communication	Usually possible	Configuration-dependent	Yes	Usually possible
Physical LAN exposure	No direct exposure by default	No direct exposure by default	Isolated by default	Directly connected
Best use	Internet access and updates	Multi-VM labs requiring internet	Isolated security labs	LAN testing
Suitability for vulnerable targets	Not preferred	Requires additional precautions	Recommended	Unsafe on ordinary networks

Why Vulnerable Labs Must Never Be Bridged

Intentionally vulnerable machines, such as Metasploitable and other deliberately insecure training systems, should never be bridged onto ordinary home, college, or office networks.

Reasons:

1. Unauthorized access: Other devices on the physical network may discover and access the vulnerable virtual machine.
2. Malware infection: Attackers or malware may exploit known vulnerabilities in the VM.

3. Network compromise: A compromised VM may be used to attack other computers connected to the same network.
4. Data theft: Attackers may gain access to sensitive information stored on the vulnerable machine or reachable network resources.
5. Accidental attacks: Security testing tools may unintentionally scan or affect devices outside the authorized testing environment.
6. Legal and ethical issues: Testing systems outside the approved scope may violate organizational policies or cybersecurity laws.

Recommended Networking Setup for Cybersecurity Labs

Kali can access the internet through NAT while communicating with the vulnerable target through host-only networking. Disable routing between these networks.

Recommended configuration:

- Kali Linux: Adapter 1 = NAT; Adapter 2 = Host-only.
- Vulnerable target: Adapter 1 = Host-only only.
- Disable IP forwarding and unnecessary network sharing between adapters.
- Keep vulnerable targets disconnected from the internet and physical LAN.

Conclusion

VM networking modes determine how virtual machines communicate with external networks, the host system, and other virtual machines. NAT provides internet access, NAT Network supports communication among multiple VMs with internet connectivity, Host-only provides an isolated testing environment, and Bridged connects a VM directly to the physical network.

For cybersecurity labs, Host-only is generally the safest choice for intentionally vulnerable machines because it reduces exposure to external networks and helps prevent accidental attacks on unauthorized systems.